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Swelling Anisotropy of Shales

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Abstract

Hydrocarbon extraction from unconventional formations requires pressurized introduction of a substantial volume of water to create permeable pathways in situ, thereby accessing optimal reservoir zones. Consequently, the interaction between shale and injected water leads to shale hydration, causing swelling and softening, which presents risks related to the stability of wellbores and the potential diminishment of induced permeability. Despite extensive investigations into the causes of shale expansion, a comprehensive evaluation of anisotropic swelling has been overlooked. Therefore, this study aims to investigate the extent of unconfined anisotropic expansion in shales with significant variations in mineralogical content, providing a comprehensive overview of the anisotropic swelling intensity and its underlying causes. Digital Image Correlation (DIC) was employed for quantitative spatial deformation analysis in well-preserved and intact samples of the Tuscaloosa Marine Shale and Eagle Ford Shale. The results indicate that unconfined swelling anisotropy can reach values exceeding 10 in highly reactive and laminated shales. While the vertical expansion of shale is primarily dependent on clay content and the presence of laminations, it was found that all structural constituents, including clay, quartz, and carbonates, highly influence the degree of horizontal swelling. Furthermore, the unconfined swelling anisotropy was discovered to be almost equally dependent on all primary minerals present. The shale samples with limited carbonate content exhibit significant expansion and are vulnerable to washout. The bidirectional expansion anisotropy was predicted for 503 shale samples of Eagle Ford and Tuscaloosa Marin Shale formations. Overall, this work sheds light on previously unexplored aspects of shale behavior, with potential implications for drilling and production optimization, as well as safe operations in unconventional reservoirs.

Keywords: Digital Image Correlation, Shale Anisotropy, Minerals Orientation, Shale Swelling, Shale-Fluid Interaction, Mineralogy

Introduction

The structural complexity and heterogeneity of shales, characterized by features such as fissility and distinct interbedded layering, govern the directional variation of mechanical strength and, consequently, deformation upon fluid exposure (Moon et al. 1984). It is well-established that shales exhibit swelling and dispersion when they come into direct contact with water, which activates reactive laminae in the formation, altering well integrity or diminishing the induced reservoir permeability (Chenevert et al. 1998; Van Oort et al. 2003; Lyu et al. 2015; Chen et al. 2003). While considerable attention has been devoted to studying and assessing shale swelling in the presence of aqueous fluids over the years, a comprehensive understanding of the anisotropic nature of shale swelling remains elusive.

Initially, during sedimentation, the orientation of the minerals within the shale structure is essentially random (Wong et al. 1996), while under the diagenetic processes and prolonged anisotropic compaction, the dominant horizontal alignment gradually develops (Loon et al. 2008; Czurda et al. 1973; Lo et al. 1978; Yuan et al. 2014). Under increasing overburden stress, the void between clay platelets diminishes, and the consequent subsurface alteration causes a further refinement of mineral alignment within the structure (Loon et al. 2008). This in-situ established orientation of minerals governs the anisotropy observed in various shale properties such as seismic velocity, thermal and hydraulic conductivity, diffusion coefficients, and swelling pressures (Loon et al., 2008). Consequently, the anisotropic arrangements of minerals in shale play a crucial role in controlling its swelling behavior.

Through the application of Digital Image Correlation (Chuprin et al. 2022, Parrikar et al. 2020), Linear Variable Differential Transformers (LVDTs) (Wakim et al. 2019), and strain gauges (Minardi et al. 2018, Dehghanpour 2012), it was revealed that the interaction of water with shales predominantly induces expansion perpendicular to the laminations. This behavior is driven by non-recoverable deformation during imbibition, such as fracturing parallel to the bedding, which is directly correlated with the concentration of reactive clay (Dehghanpour et al., 2013; Chuprin et al., 2022). This phenomenon leads to increased permeability along the bedding planes (Kwon et al. 2004), highlighting the significant influence of shale laminations on directional expansion. The horizontal swelling, however, was reported to be less prominent, possibly due to the preferential orientation of clay platelets.

While previous studies addressed directional swelling behavior, a systematic investigation into swelling anisotropy and its relationship with mineralogical composition remains lacking. Therefore, this study seeks to address this gap by providing a comprehensive examination of the intricate anisotropic swelling behavior of shales with varying mineralogical compositions. By offering a novel perspective on the anisotropic properties of shale, this work contributes to a deeper understanding of shale-fluid interaction dynamics relevant to hydrocarbon extraction processes.

Materials and Methodology

This section outlines the Digital Image Correlation (DIC) experimental procedures, including detailed material descriptions, preparation, and measurement settings. The shale samples tested comprise Upper Cretaceous Tuscaloosa Marine Shale (TMS) and Upper Cretaceous Eagle Ford (EF) shales, both of which are known for their pronounced fissility and well-defined layering. The mineralogical composition of 503 shale samples is illustrated in the ternary plot in Figure 1. From this dataset, nine representative specimens, six TMS and three EF, were selected for detailed quantitative and qualitative analysis to evaluate deformation trends and the influence of mineralogical content on swelling anisotropy. The composition of tested TMS samples ranges from 53 to 9 wt.% in clay, 87 to 36.5 wt.% in quartz, and 15 to 1% in carbonates. In contrast, the EF shale samples vary from 27 to 1 wt.% in clay, 25 to 3 wt.% in quartz, and 91 to 41 wt.% in carbonates. Additionally, DIC swelling data from the 7th TMS specimen, previously analyzed by Chuprin et al. (2022), was incorporated to include a case with a clay content exceeding 60 wt.%.

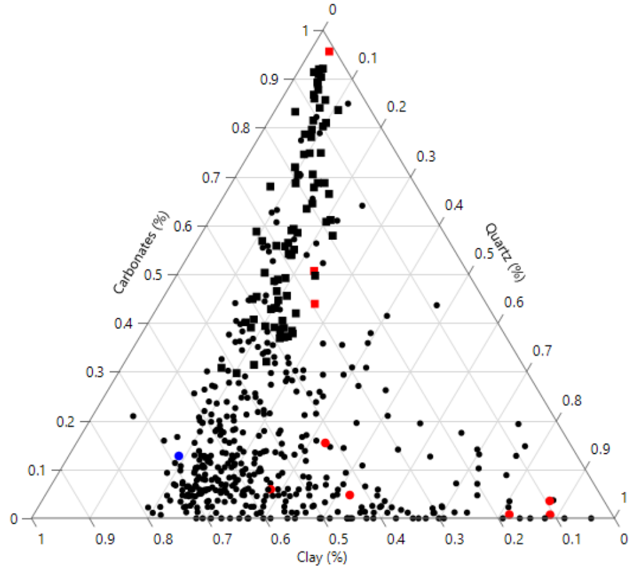

Figure 1—The mineralogical composition of 503 investigated shale samples (420 of TMS; 83 of EF); Red circles are the selected samples of TMS; Red squares are the selected samples of EF; Blue circle is a referenced sample from (Chuprin et al. 2022)

Particular attention was given to a specific TMS sample that exhibited distinctive erosion from coring procedures at the well site, indicating an unstable shale region prone to washout. The Figure 2 showcases 360° images of the washout section, which was divided into two distinct regions: an extensively altered argillaceous upper segment and a quartz-dominated lower segment. For this study, the upper eroded portion was selected as representative of a compromised wellbore section, providing insight into shale behavior under destabilizing conditions.

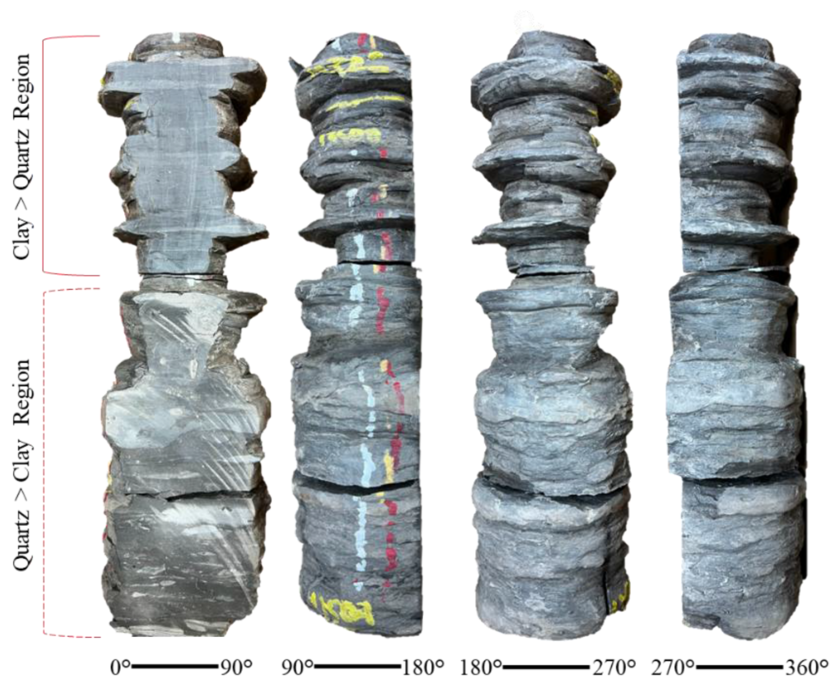

Figure 2—Washout section of the Tuscaloosa Marine Shale (TMS)

Digital Image Correlation

Digital Image Correlation is an optical, non-contact technique that enables full-field spatial strain measurements across the surface of an object. This technique offers significant advantages over stain gauges and linear swelling meters, as it enables assessment of spatial strain variations and allows the investigation of deformation in intact samples without the need for pulverization. The sample preparation for this experiment involved extracting cuboidal samples from the core at the depths of interest. The sample's surface was polished, and the abraded particles were removed for better paint adhesion. The surface of interest was painted, creating a black-on-white speckle pattern. The formed pattern was divided into subsets of 45 pixels each, with a step size of 12 pixels. The displacement of the speckles within the subsets was analyzed and measured using DIC software (Aramis 2019). The samples were immersed in DI water until the semi-stabilized state was reached. Strain measurements were evaluated at 4 days to normalize the strain development for all samples. The longitudinal (vertical) and transverse (horizontal) strains were calculated as a function of time, and the strain maps for several time stages were presented. Whereas the anisotropy of the formation was calculated as:

$$I_s = \frac{\max_t f\,(\bar{\varepsilon}_{yy})}{\max_t f\,(\bar{\varepsilon}_{xx})} \tag{1}$$

Where, $\bar{\varepsilon}_{yy}$ and $\bar{\varepsilon}_{xx}$ are the measured average longitudinal and transverse strains at the final time step. The unconfined anisotropy indexes of <1, =1, >1 represent predominant shale expansion in the horizontal direction, isotropic expansion, and predominant shale deformation in the vertical direction relative to the bedding, respectively.

Results

This section presents the results of five DIC tests conducted on EF and TMS shales. Deformation maps illustrating the evolution of vertical and horizontal strains are selectively displayed to emphasize notable irregularities and anisotropic swelling behavior. Figure 3 shows the direction-oriented average strains along with the corresponding horizontal and vertical strain maps for three TMS samples. The development of vertical and horizontal strains within the washout TMS sample revealed a consistent pattern of directional deformation, with vertical swelling reaching up to 11 times the magnitude of horizontal expansion. Initial swelling was triggered through the activation of lamination planes, which exhibited continuous vertical expansion. By time step t3, the matrix swelling and activation of secondary fractures were recorded. In contrast, a second TMS sample was hydrated for 21 days to explore the extended shale-water interaction. After four days, no significant displacement was observed, but continuous expansion persisted within previously activated regions. In this case, swelling remained concentrated along the initially active laminations, with no new zones of deformation emerging. The total vertical swelling in this specimen exceeded a factor of 12 relative to horizontal expansion. A third sample, representative of a low-clay TMS shale, exhibited markedly different behavior in the absence of distinct reactive laminae. The expected exponential expansion was not observed owing to the limited swelling potential. Instead, irregular fractures began to form around day 3, contributing to both horizontal and vertical expansion, influenced by their non-laminar orientation.

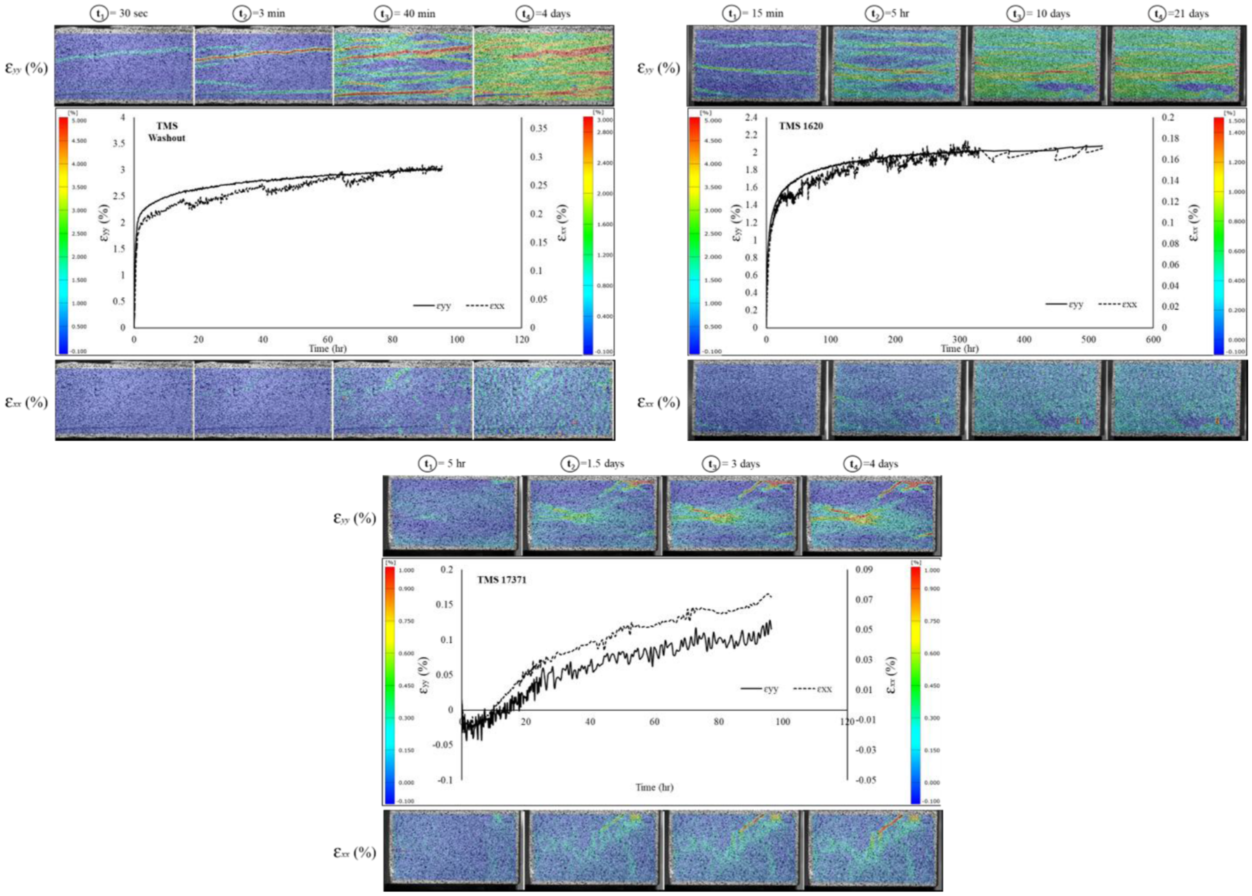

Figure 3—Temporal vertical and horizontal strain progressions and the resulting DIC horizontal and vertical strain maps for three TMS samples. From left to right: TMS washout sample, TMS sample with clay content ~41%, TMS shale with ~ 80% quartz.

The Figure 4 displays the temporal direction-oriented average strain progressions and the resulting horizontal and vertical strain maps for two EF samples. The EF sample with limited clay content exhibits limited swelling. Due to the small-scale vertical strain fluctuations, the moving average was provided for this specimen. A similar pattern of strain evolution in vertical and horizontal directions was observed. The horizontal strain was dominant in this case, whereas the spatial deformation is distributed across the surface in both directions without pronounced reactive regions. The EF samples with high clay content display distinct behavior compared to TMS. Instead of specific lamination activation on the surface, EF undergoes mostly cumulative swelling, with a single high swelling region detected. The cumulative expansion covers the entire surface, including upper threshold expansion areas. This expansion continues throughout the experiment, resulting in an anisotropy exceeding 10. The intensity of swelling differs, with more pronounced exponential expansion in the vertical direction compared to the horizontal direction.

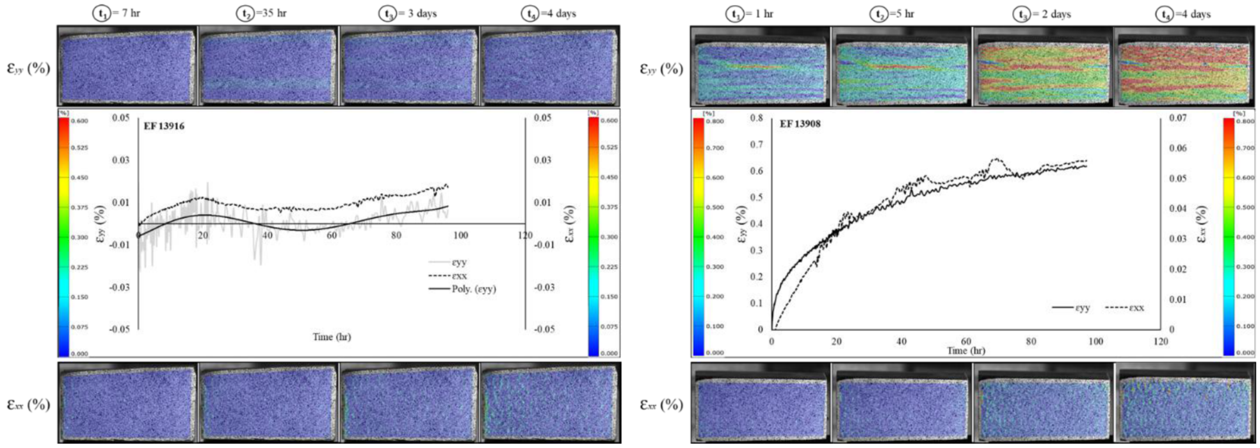

Figure 4—Temporal vertical and horizontal strain progressions and the resulting DIC horizontal and vertical strain maps for two EF samples. From left to right: EF sample with ~1% clay, EF sample with ~28% clay

Based on the results obtained from the DIC, swelling in both horizontal and vertical directions was recorded at 4 days, marking the experimental endpoint for the majority of the tested samples. The total horizontal and vertical swelling, representing the average displacement from all shale subsets on the surface, was plotted against clay, quartz, and carbonate concentrations, as shown in Figure 5.

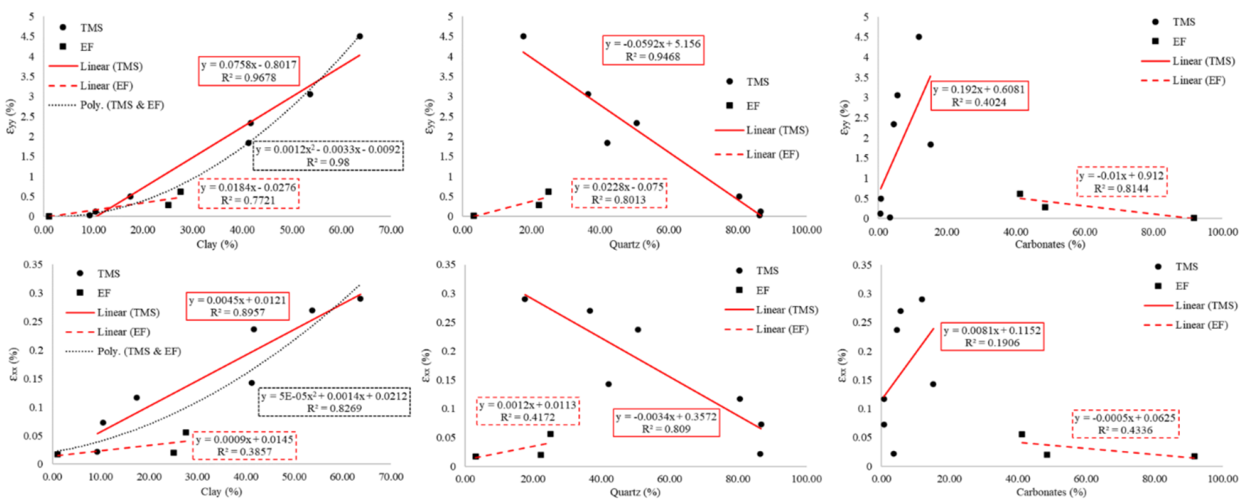

Figure 5—Measured vertical and horizontal strains of both TMS and EF shale samples as a function from left to right: clay, quartz, carbonates %.

The total average vertical strain in all tested shale types was found to be strongly dependent on clay concentration. In TMS, vertical strain exhibited a strong positive linear correlation with clay content (R^2 = 0.98) and a strong negative correlation with quartz content (R^2 = 0.92), indicating that TMS swelling increases as clay replaces quartz. Calcite exhibited a negligible correlation with vertical strain, suggesting a limited role in vertical deformation (R^2 = 0.2). In contrast, horizontal strain demonstrated greater sensitivity to mineralogical heterogeneity, with R^2 values of 0.76 for clay, 0.62 for quartz, and 0.02 for calcite. This implies that while the clay is a primary driver of transverse deformation, horizontal swelling is also influenced by other factors, such as pre-existing structural discontinuities or cementation within the shale matrix. Additionally, the reduced horizontal orientation of the clay pellets likely contributed to the lower horizontal strain response.

In EF shale, mineralogy-dependent deformation patterns in the vertical direction exhibited a positive correlation between vertical expansion and clay and quartz wt.% (R^2 = 0.77 and 0.8, respectively).

Conversely, a negative correlation was observed between vertical deformation and carbonates, primarily attributed to the inverse relationship between calcite and clay concentrations. In the horizontal direction, the correlations between strain and individual mineral components were notably weaker, indicating a diminished dependence on clay content alone. This suggests that horizontal deformation in EF shale is more strongly influenced by the complex interplay of heterogeneous mineral constituents and structural variability, rather than by a single mineral phase.

The linear relationship between unconfined swelling anisotropy and mineralogical composition was investigated for both TMS and EF shales, as shown in Figure 6. Unconfined swelling anisotropy was found to have a strong linear dependence on the concentrations of all major mineral phases, indicating that bulk deformation behavior is closely tied to mineralogy. These trends closely mirror the correlations observed in direction-specific swelling, but are more pronounced. In TMS, anisotropy is primarily governed by the inverse relationship between clay and quartz content, whereas EF displays the opposite trend. Notably, the coefficient of determination for the unconfined swelling correlations is substantially higher than those obtained from individual direction strain analyses, suggesting that bulk anisotropic deformation provides a more integrated measure of the shale's mineralogical influence.

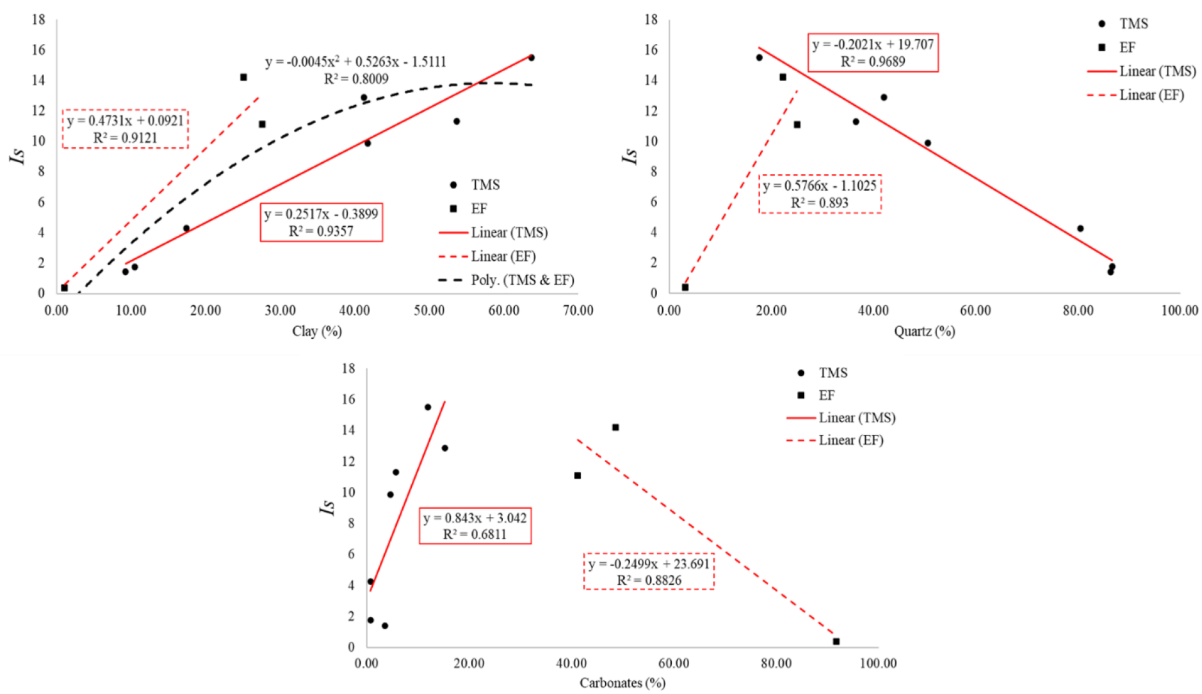

Figure 6—Experimental swelling anisotropy of both TMS and EF shale samples as a function from left to right: clay, quartz, and carbonates %

Utilizing the established polynomial correlations of clay variable to vertical and horizontal strains, Figure, (a), the vertical and horizontal swelling was predicted for the grouped shales using eq. 2 and eq.3.

$$\varepsilon_{yy} = 0.0012x^2 - 0.0033x - 0.0092 \tag{2}$$

$$\varepsilon_{xx} = 5 * 10^{-5}x^2 + 0.0014x + 0.0212 \tag{3}$$

The results presented in Figure 7, demonstrate a high degree of agreement between the predicted and experimental values for both vertical and horizontal swelling, confirming the accuracy of the established polynomial models.

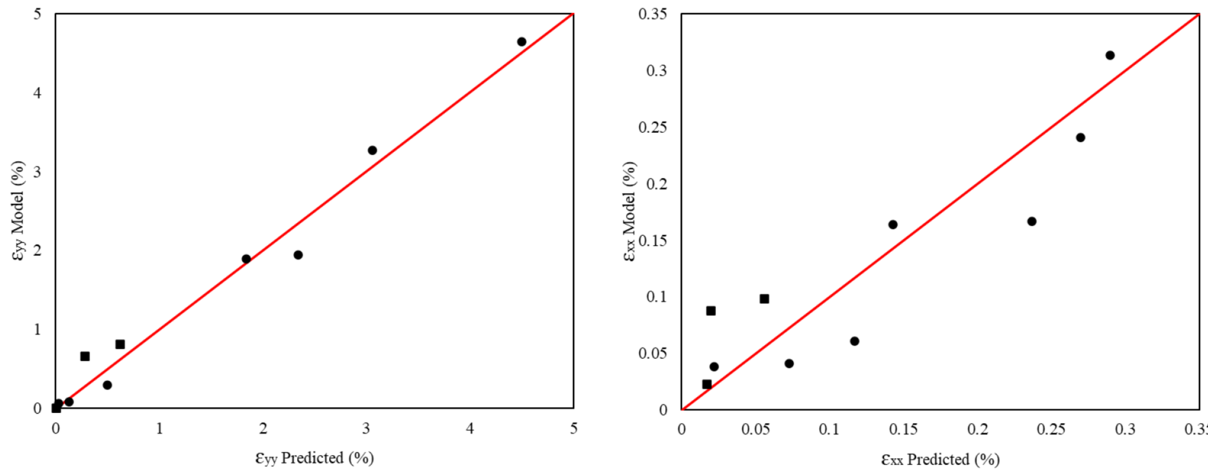

Figure 7—Predicted vertical and horizontal strains for both TMS and EF tested shale specimens.

The vertical swelling for the dataset of 503 shale samples was predicted and demonstrated in Figure 8. The maximum potential vertical swelling, approximately 7%, was associated with the highest recorded clay concentration and attributed to the formation of TMS. Based on the input data, the predicted swelling distributions are lognormal for quartz and exponential for calcite. For quartz, the highest vertical swelling values were observed within the concentration range of 15-25 wt.%, while calcite-related swelling was most pronounced at concentrations below 4 wt%. These trends highlight the differing roles of siliceous and carbonate minerals in controlling the extent of vertical deformation in TMS shale.

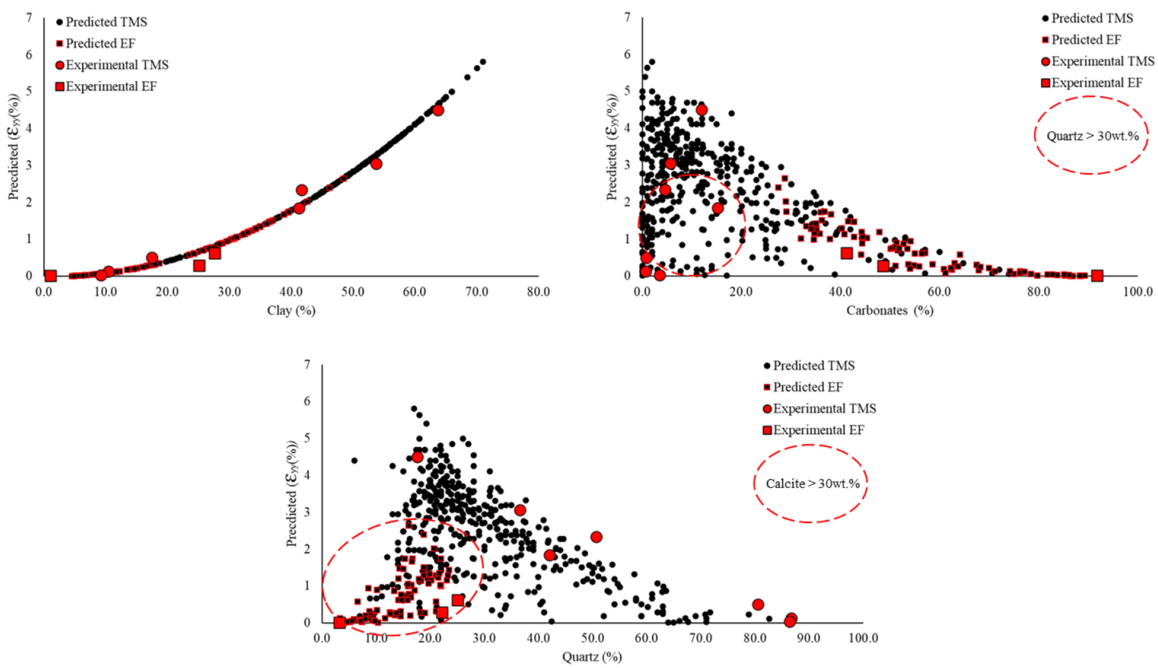

Figure 8—Predicted vertical strain for the total of 503 shale specimens representing TMS and EF as a function of clay, carbonates, and quartz % (from left to right)

Utilizing validated vertical and horizontal prediction models, an attempt to forecast swelling anisotropy was made for 493 samples based on raw mineralogical data from both TMS and EF shales, Figure 9. Based on the results, the experimental data for unconfined swelling can be divided into two regions: clay content under 30wt.% and clay content above 30wt.%. The first region exhibited an exponential growth in swelling anisotropy as the clay concentration rose, whereas the second region showed a significant decrease in slope, indicating a plateauing effect. The analysis also revealed mineral-specific trends in anisotropy. The highest swelling anisotropy occurred when calcite content was between 0-10wt.%, with an evident decline in anisotropy observed at higher calcite concentrations. This occurred likely due to its non—swelling nature and cementing properties, restricting directional swelling in larger quantities. For quartz, the highest anisotropy is found when its concentration falls within the range of 20-30% by weight.

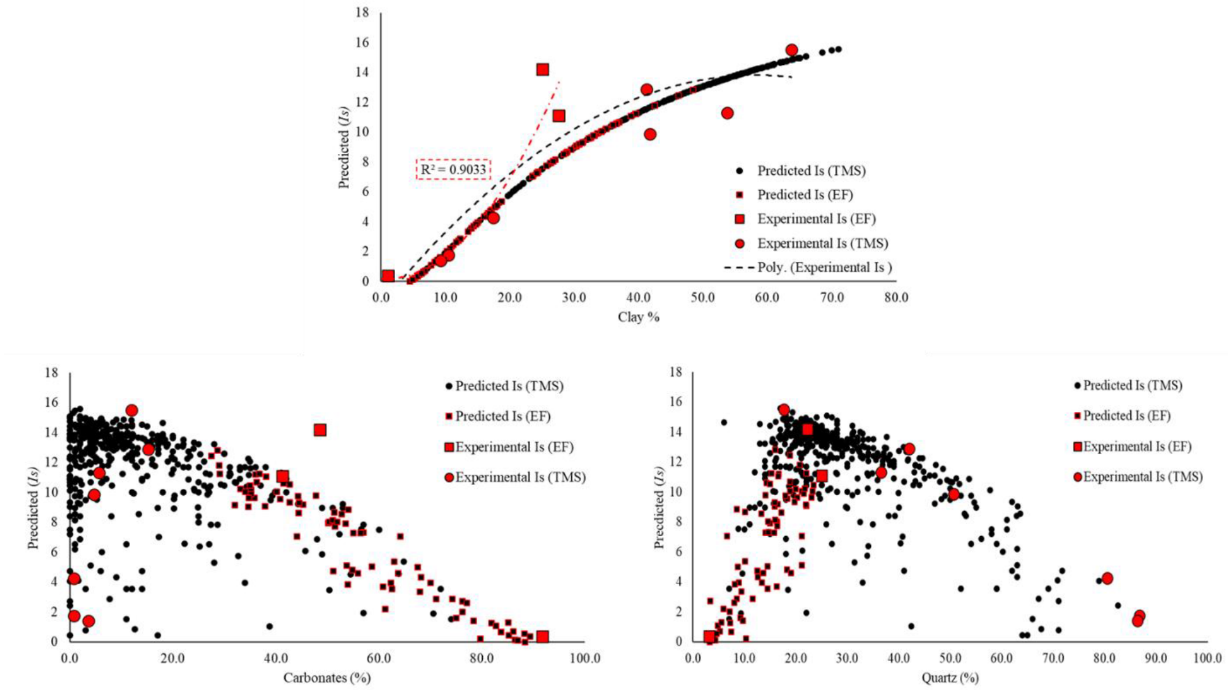

Figure 9—Predicted swelling anisotropy for the total of 503 shale specimens representing TMS and EF as a function of clay, carbonates, and quartz% % (from left to right)

Discussion and Analysis

Unconfined swelling anisotropy was investigated using Digital Image Correlation, which enabled the precise measurement of total vertical and horizontal strains evolution and mapping of localized strain development on shale surfaces. The results show a significant increase in swelling anisotropy with the rise of clay concentration in both TMS and EF shales. The greatest swelling anisotropy was observed in samples containing minimal carbonate content and quartz concentrations between 20% and 30% by weight.

The vertical expansion was found to be more significantly affected by the presence of clay than the expansion in the horizontal direction. The investigation revealed a highly pronounced correlation between the presence of clay and vertical swelling, with an R^2 value of 98%. In contrast, the unconfined horizontal deformation is influenced by factors beyond the mere presence of clay. This phenomenon may be attributed to variations in the proportion of clay oriented sub-vertically, laminations, and constraints imposed by carbonate cementation.

Both TMS and EF shales exhibited higher overall swelling and anisotropy in the presence of reduced carbonate content. Vertical expansion was found to be less affected by calcite cementation, since the highest strain occurs within the clay-rich reactive layers and along activated laminae. This results in upward-directed swelling pressure that lifts the overlying rock section. In contrast, horizontal swelling is more constrained, likely due to the limited presence of vertical fractures and resistance imposed by coarse-grained carbonate cementation, which restricts lateral deformation. This distinction underscores the complexity of shale deformation and its dependence on multiple geological factors.

The swelling experiments were conducted under atmospheric conditions. While the vertical expansion of shale is significantly greater than the horizontal expansion in the unconfined state, in situ confined pressure might suppress this swelling by several folds (Minardi et al. 2018; Wakem et al. 2019; Ewy et al. 2010; Huang et al. 2023; Chenevert et al. 1992). Previous studies show that shale samples with a concentration of clay above 50% may exhibit reduced swelling under confinement (Huang et al. 2023; Ewy et al. 2010; Minardi et al. 2018). Even though vertical expansion may be significantly limited in situ for most shale formations, horizontal expansion still occurs (Ewy et al. 2022). This behavior may be attributed to a reduction in compressive strength and the formation of microcracks (Li et al., 2023), which act as a counterforce to the vertical confining pressure during fluid inhibition (Huang et al., 2023).

As demonstrated by the DIC experiments, most samples reach a swelling pseudo-steady state within 4 days, which coincides with previously reported results (Chuprin et al., 2022; Parrikar et al., 2022). However, to investigate the deformation under more prolonged submersion, a TMS sample with a clay content of 41 wt.% was monitored over a 21-day period. The results indicate an increase in vertical and horizontal strain by 11% and 18%, respectively, from 4 to 21 days of shale interaction with water. This suggests that horizontal expansion continues at a relatively higher rate during the later stages of water interaction, signifying that long-term exposure may enhance anisotropic deformation, particularly in the transverse direction.

The washout section of the TMS exhibited substantial deformation upon contact with water, which is attributable to the high clay content within its structure. However, while clay plays a significant role in shale disintegration, the root cause of washout appears to be the compounding effect of several factors. Whereas a broad spectrum of unconventional formations prone to a controllable washout as a consequence of the fissile nature (Ghajari et al. 2013), some of the shales exhibited this alteration to a significant degree, requiring a subjective solution approach (Chenevert et al. 1970). Analysis of the mineralogical composition revealed a limited presence of carbonates in the tested sample, which is consistent with the previously reported cases (Cheenvert et al. 1970; Guo et al. 2006). The lack of calcium carbonate precipitation between the grains results in structural integrity flaws, making the shale less resistant to disintegration and erosion. The study of microstructure showed abundant microfractures in such regions (Guo et al. 2006). Thus, these pathways to the matrix increase the contact surface area between the clays and the injected water. Since natural cementation is absent, this part of the formation lacks cohesion, leading to a significant decrease in strength when compared to well-cemented carbonate formations. The diminished strength and cohesion make the formation more susceptible to mechanical deformation. Consequently, the combined influence of swelling clays and the nature of quartz leads to instability of the wellbore walls and mechanical abrasion, inducing wellbore enlargement and washout phenomenon.

CONCLUSIONS

This work presented a comprehensive analysis of unconfined anisotropic swelling in TMS and EF shales using Digital Image Correlation (DIC). The Digital Image correlation technique enabled spatially resolved quantification of vertical and horizontal strain evolution, as well as the generation of temporal displacement maps during shale-water interaction. Mineralogical analysis was conducted to investigate the relationship between shale composition, specifically discrete mineralogical phases, and direction-dependent deformation. Based on the experimental results, the following conclusions can be drawn:

1. The maximum experimental unconfined swelling anisotropy of the TMS and EF shales was recorded at 12.8 for TMS and 14.2 for EF. In contrast, the maximum modeled anisotropy approached 16 in TMS shale with clay concentration above 60%.
2. The peak swelling anisotropy was associated with minimal calcite content, moderate quartz presence, and a dominant clay fraction. This combination appears to enhance vertical deformation during water imbibition.

3. The unconfined vertical expansion of shale is directly related to clay concentration, with an R^2 of 0.98. In contrast, the horizontal swelling is more variable, influenced by clay orientation, compositional heterogeneity, and the presence of coarse-grained cementation, resulting in a lower R^2 of 0.82.
4. Shales containing high clay content and low calcite are particularly susceptible to significant swelling, elevated anisotropy, and associated risks such as borehole washout.

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